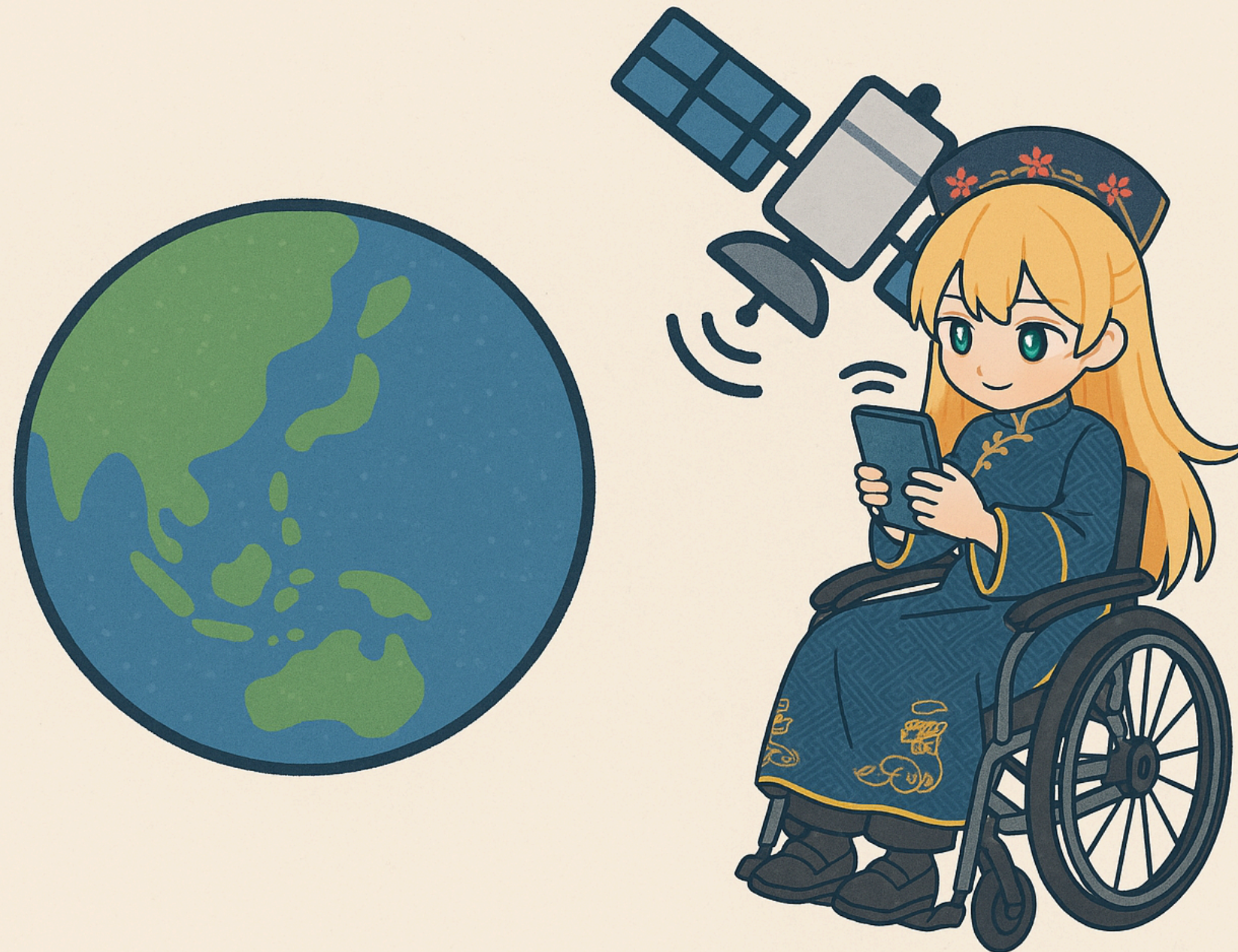


RELATIVITY IN GPS

HOW DO SATELLITES MEASURE THE LOCATION OF
SMARTPHONES?



TODAY'S TOPIC

-  How does GNSS like GPS measure position?
 - Case without considering relativistic effects (Geometric Optics)
 - Effects of Special Relativity
 - Effects of General Relativity
-  You'll understand what's happening when your smartphone shows your location on  map apps!

ASSUMPTIONS

- Earth is a sphere
 - Earth does not rotate
 - Earth is electrically neutral
 - Earth's mass distribution is uniform
 - Earth has no atmosphere (it's a vacuum)
 - No other celestial bodies (like the Moon) exist
-
- These assumptions are far from reality, but they simplify the problem!
 - Surprisingly, this approximation works well

TECHNICAL TERMS

- GNSS (Global Navigation Satellite System) is the general term for satellite-based positioning systems
 -  GPS (Global Positioning System) is one type of GNSS
 - Others include  GLONASS,  Galileo,  BeiDou,  Michibiki
- Today, we'll refer to it as GPS, which is most familiar to Japanese people
 - It's like calling all game consoles "Famicom" - technically not correct, but commonly used

POSITION MEASUREMENT USING GEOMETRIC OPTICS

SIMPLE EXAMPLE OF DISTANCE MEASUREMENT

- Starting from point A at time $t = 0$ (s) with velocity $v = 10$ (m/s) towards point B
- Arriving at point B at time $t = 10$ (s)
- What is the distance between A and B ?

$$||AB|| = 10 \text{ (m/s)} \times 10 \text{ (s)} = 100\text{m}$$

- **Distance between two points can be calculated as speed $\times$ time!**

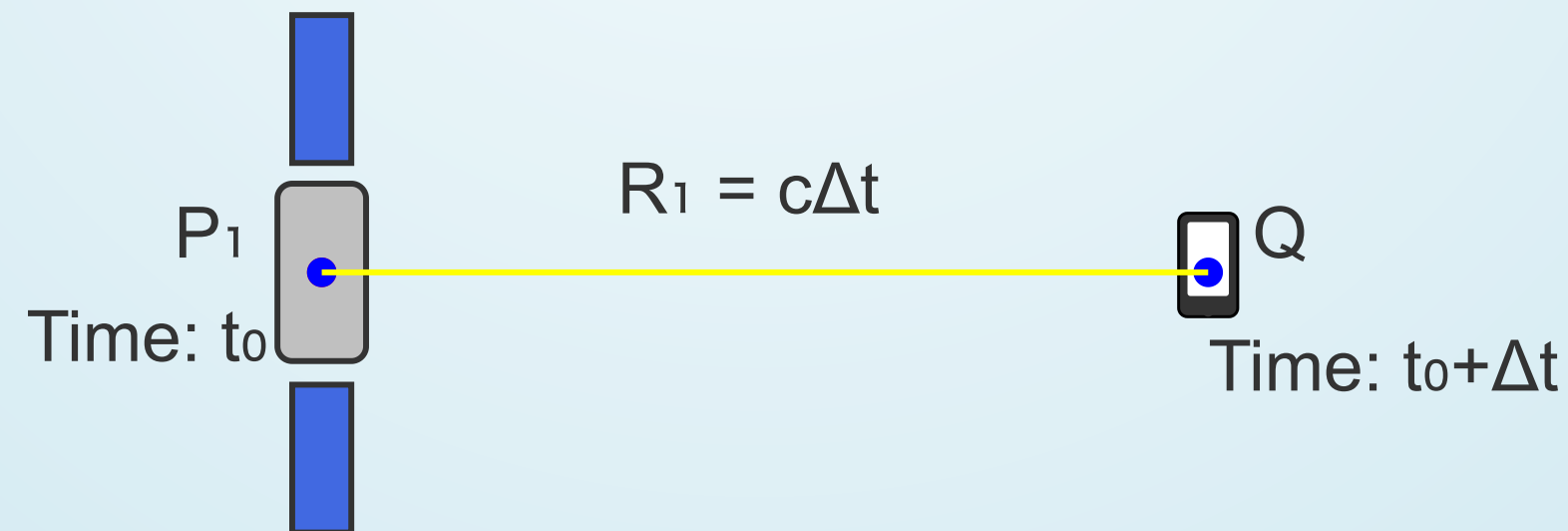
DISTANCE MEASUREMENT USING LIGHT

- Speed of light: $c = 3.0 \times 10^8 (\text{m/s})$
- Satellite transmits electromagnetic waves from point P at time t_0
- Smartphone receives the waves at point Q at time $t_0 + \Delta t$

$$||PQ|| = c\Delta t$$

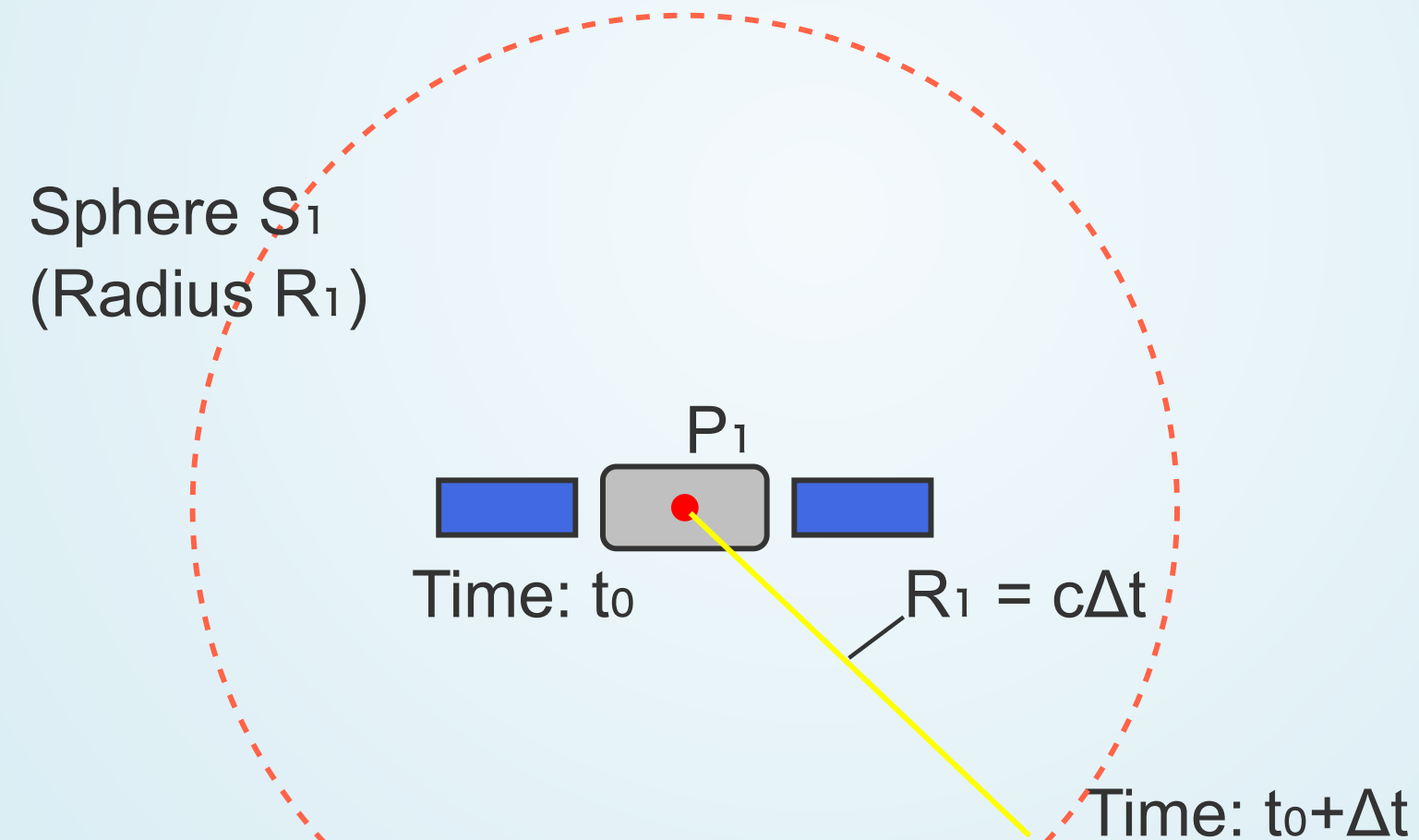
POSITIONING WITH ONE SATELLITE

- Satellite transmits radio waves from point P_1 at time t_0 , smartphone receives it after Δt
- Distance between points P_1 and Q is $R_1 = c\Delta t$



WHERE IS THE SMARTPHONE?

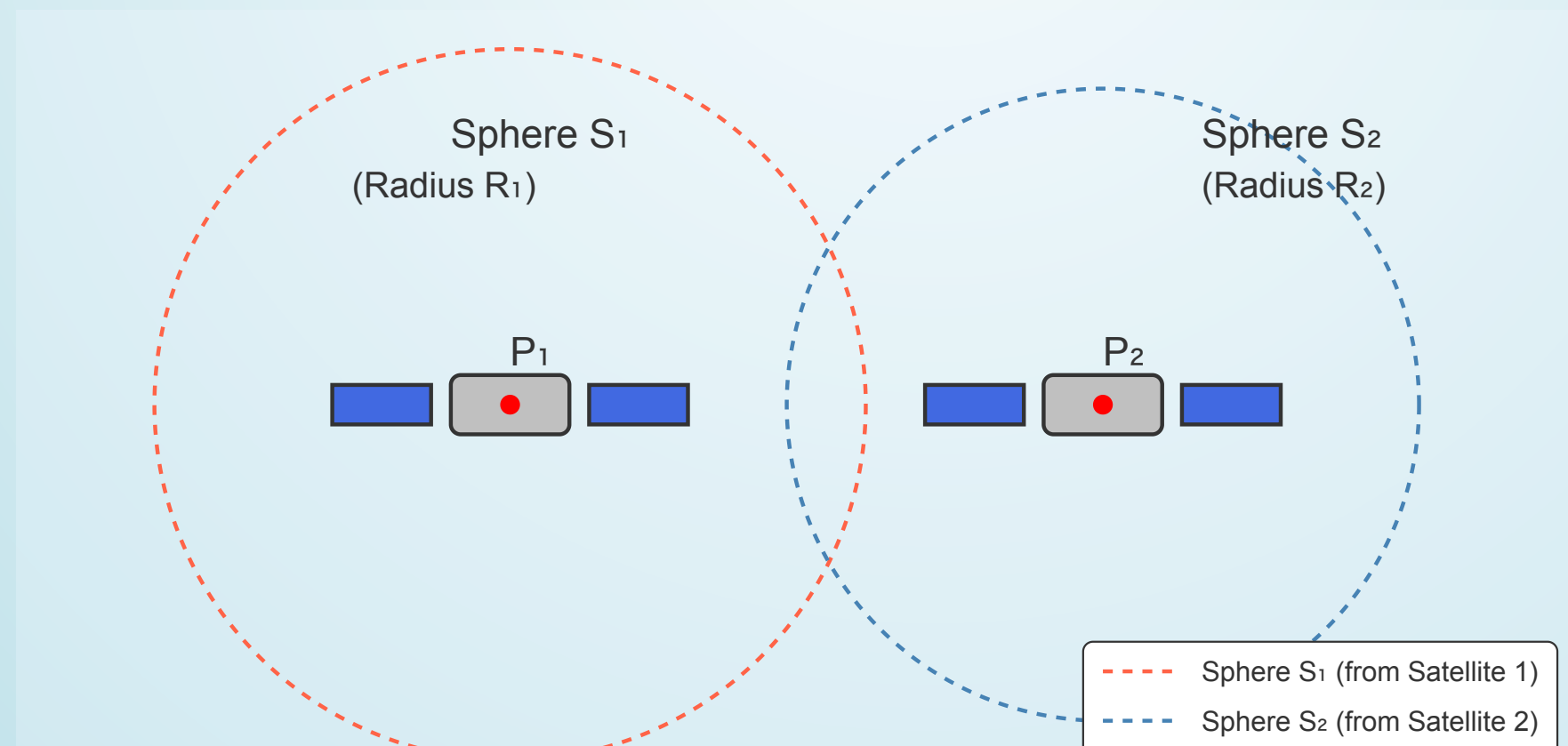
- It's somewhere on the surface of sphere S_1 with center P_1 and radius R_1 !



The smartphone is somewhere on the surface of Sphere S_1 !

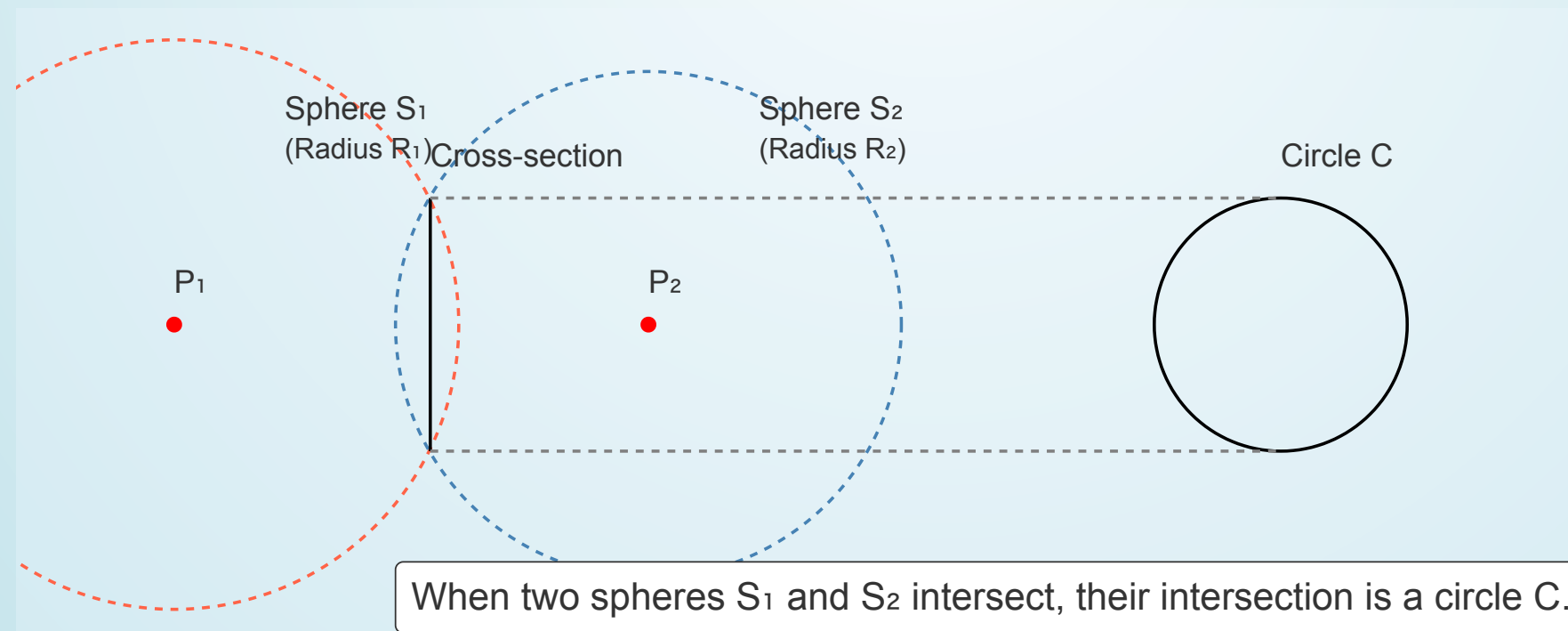
POSITIONING WITH TWO SATELLITES

- Similarly measuring distance from two satellites
 - Smartphone is somewhere on the surface of sphere S_1
 - And also on the surface of sphere S_2



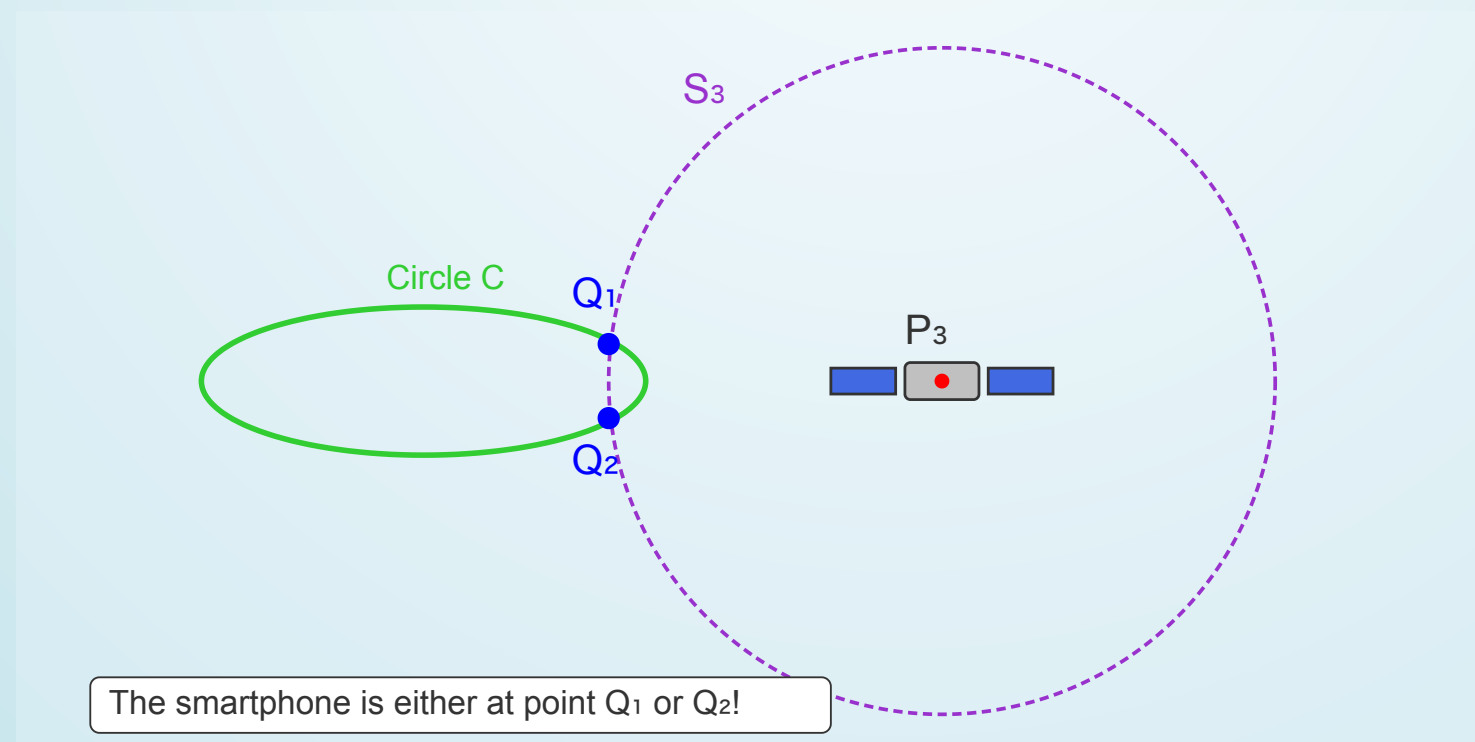
NARROWING DOWN THE SMARTPHONE'S POSITION

- The intersection of two spheres is a circle
- It's somewhere on the circumference of circle C !
- **By increasing the number of satellites, we can narrow down the position!**



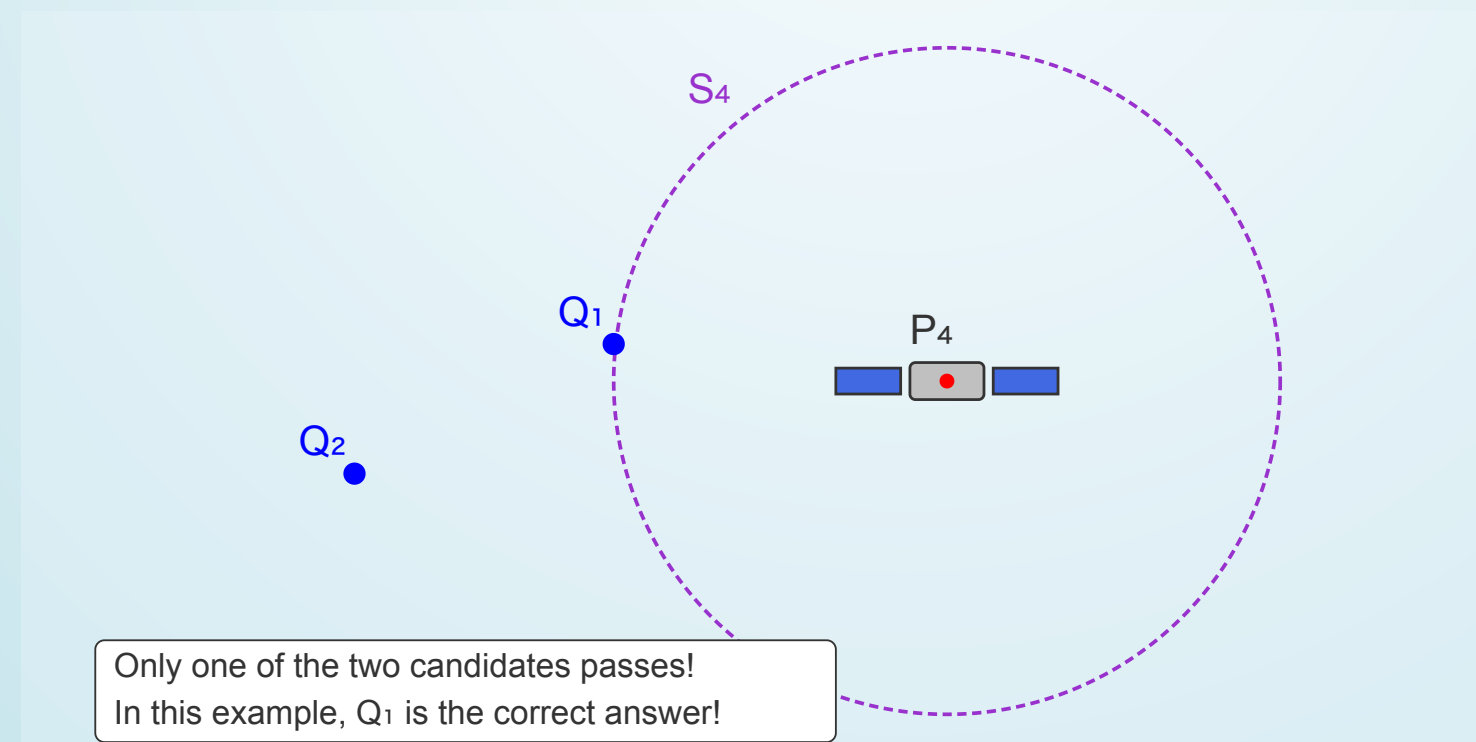
POSITIONING WITH THREE SATELLITES

- Getting sphere S_3 from the third satellite
- We know it's somewhere on the circumference of circle C , which intersects at two points
- **Now we're down to two possible points!**



POSITIONING WITH FOUR SATELLITES

- Getting sphere S_4 from the fourth satellite
- **With distance measurements from four satellites, we can determine the smartphone's position Q !**



SUMMARY

- Receiving signals from one GPS satellite, we know the smartphone is on the surface of a sphere
- Receiving signals from four satellites, we can determine the position
- **We can determine the smartphone's position from the intersection of spheres!**

ADDITIONAL NOTES

* In practice, we treat the Earth's surface as another sphere, and perform positioning with three satellites * For aircraft, positioning from the fourth satellite is important!

EFFECTS OF SPECIAL RELATIVITY

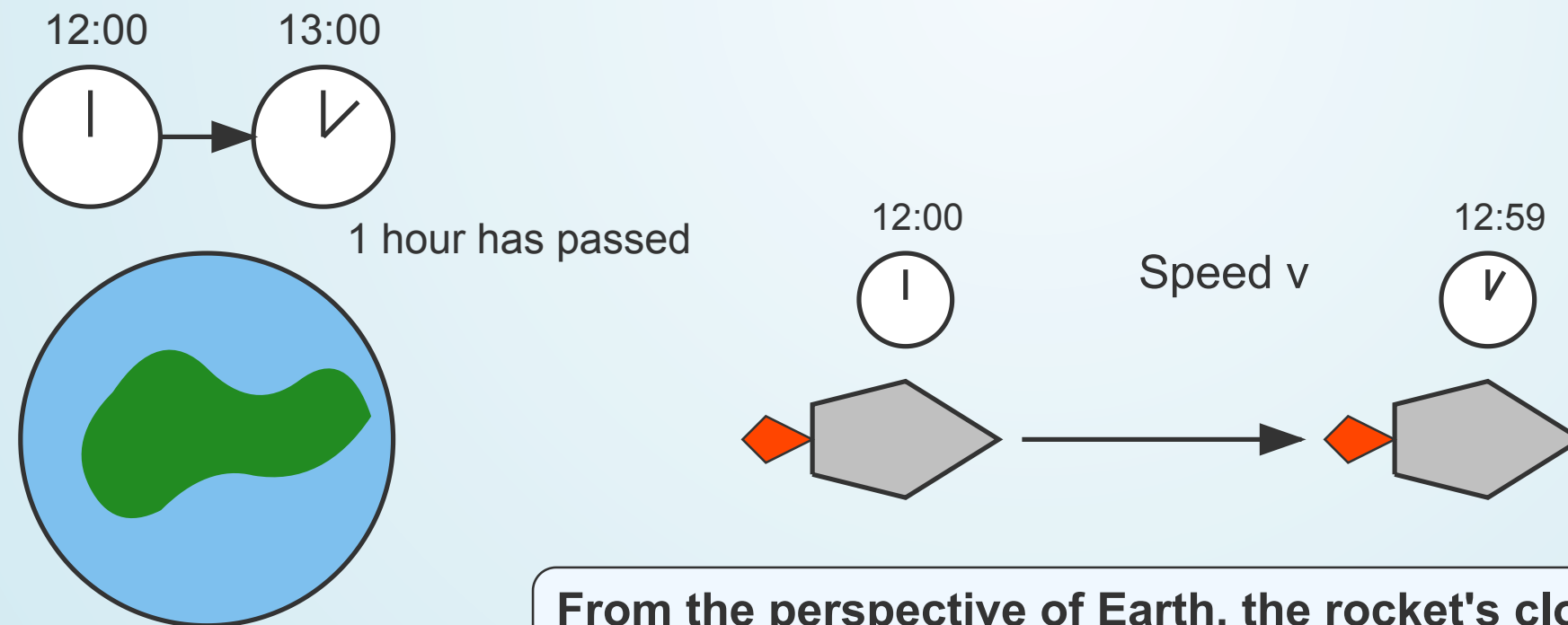
IMPORTANT CLAIM OF SPECIAL RELATIVITY

- Objects moving at high speeds appear to time slower to observers at rest
 - This is a highly simplified explanation!



EXAMPLE OF SPECIAL RELATIVITY

- Consider a rocket moving at speed v relative to Earth
- **From Earth's perspective**, the rocket clock appears to run slower!



From the perspective of Earth, the rocket's clock is

🚀 The rocket's clock is

🌍 The Earth's clock is

slower!

LORENTZ FACTOR

- When the rocket clock has advanced t' seconds, the Earth clock has advanced t seconds
 - This γ is called the Lorentz factor

$$t' = \frac{t}{\sqrt{1 - v^2/c^2}} = \gamma t$$

SIMPLE EXAMPLE OF TIME DILATION

- Let's consider a rocket with speed $v = 4.2 \times 10^7 \text{ (m/s)}$
- The Earth clock advances by 1 second while the rocket clock advances by 0.99 seconds
 - From Earth's perspective, **the rocket clock is 0.01 seconds ahead**

$$\Delta t = 1 - \frac{t'}{\gamma} = 1 - \sqrt{1 - \frac{(4.2 \times 10^7)^2}{(3.0 \times 10^8)^2}} \simeq 0.01(\text{s})$$

EFFECTS OF SPECIAL RELATIVITY ON TIME

- Speed of GPS satellite
 - Height of GPS satellite is about 20,000 km
 - Speed of GPS satellite is about 3.9 km/s

$$\Delta t_{\text{sec}} = 1 - \frac{t'}{\gamma} \simeq 1 - 8.5 \times 10^{-11}$$

- The Earth clock advances by 1 second while the GPS satellite clock is 8.5×10^{-11} seconds ahead

DAILY TIME DIFFERENCE

- 1 day is $24(\text{h}) \times 60(\text{min}) \times 60(\text{s}) = 86400(\text{s})$
 - Daily GPS satellite clock delay:

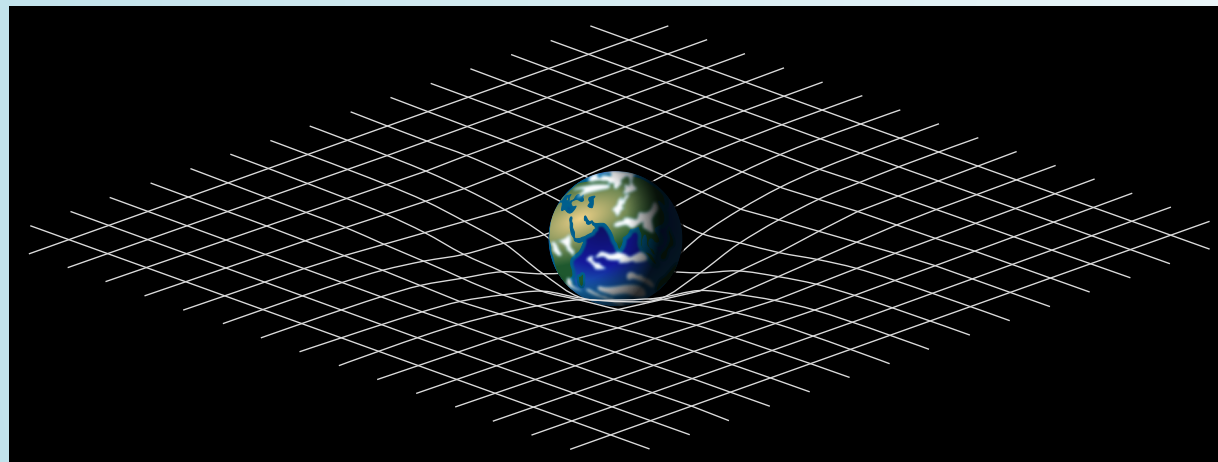
$$\Delta t_{\text{day}} = 86400(\text{s}) \times 8.5 \times 10^{-11} \simeq 7.3(\mu\text{s})$$

- $c\Delta t_{\text{day}} \simeq 2.2(\text{km})$ so **a significant time difference occurs in distance measurement!**
- GPS positioning requires correction for this time difference!

EFFECTS OF GENERAL RELATIVITY

CLAIM OF GENERAL RELATIVITY

- Mass creates spacetime distortion
- This distortion is gravity
- Time passes slower in gravity



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HOW GRAVITY AFFECTS TIME

- Stronger gravity makes time pass slower
 - Weaker gravity makes time pass faster
-
- Earth's surface is closer to Earth's center, so gravity is stronger, making the clock slower
 - GPS satellite is at an altitude of 20,000 km, so gravity is weaker, making the clock faster

SCHWARZSCHILD METRIC

- The simplest metric to represent spacetime distortion (gravity)
 - r is the distance from the body's center
- **Earth's gravity can be approximated by the Schwarzschild metric**

$$ds^2 = - \left(1 - \frac{2GM}{c^2 r} \right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r} \right)^{-1} dr^2 \\ + r^2 (d\theta^2 + \sin^2 \theta d\phi^2)$$

WHY IS EARTH'S GRAVITY APPROXIMATED BY THE SCHWARZSCHILD METRIC?

- Assumptions of the Schwarzschild metric
 - Body is spherically symmetric and stationary
 - Body does not rotate
 - Body is electrically neutral
- Today's Earth satisfies these conditions!
 - The actual Earth has a negligible effect due to this approximation

FURTHER APPROXIMATION OF SCHWARZSCHILD METRIC

- The speed of the smartphone measuring position is very slow compared to the speed of light, so we can consider it almost stationary!
 - **We can ignore the spatial components of the Schwarzschild metric!**

$$ds^2 = - \left(1 - \frac{2GM}{c^2 r} \right) c^2 dt^2$$

- Time dilation is only due to gravity!

TIME DILATION DUE TO EARTH'S GRAVITY

$$d\tau = -\frac{ds^2}{c^2} = \sqrt{1 - \frac{2GM}{c^2 r}} dt \simeq \left(1 - \frac{GM}{c^2 r}\right) dt$$

- Here r is the distance from Earth's center
- The Earth's center advances by 1 second while the point at distance r advances by the right-hand side
 - Faster in stronger gravity

TIME DIFFERENCE BETWEEN EARTH AND SATELLITE

- Let's denote the Earth's radius as R
- Let's denote the altitude of the GPS satellite as h
 - Distance from Earth's center of the GPS satellite is $R + h$
- Time difference between the Earth clock and the GPS satellite clock:

$$\Delta t = \left(1 - \frac{GM}{c^2(R + h)} \right) - \left(1 - \frac{GM}{c^2 R} \right)$$

VALUES USED IN CALCULATION

- Substitute the following values:
 - $G = 6.674 \times 10^{-11} (\text{m}^3 \text{kg}^{-1} \text{s}^{-2})$
 - $M = 5.972 \times 10^{24} (\text{kg})$
 - $R = 6.378 \times 10^6 (\text{m})$
 - $h = 2.0 \times 10^7 (\text{m})$
 - $c = 3.0 \times 10^8 (\text{m/s})$

CALCULATION RESULT

- Daily time difference between the GPS satellite clock and the Earth clock:

$$\Delta t_{\text{day}} \simeq 45.5(\mu\text{s})$$

- 1 day, the GPS satellite clock is $45.5 \mu\text{s}$ ahead of the Earth clock
- Larger than the time delay due to special relativity!

TOTAL TIME DIFFERENCE

- Time delay due to special relativity is $-7.3\mu s$ (delay)
- Time advance due to general relativity is $+45.5\mu s$ (advance)

- Adding both:

$$\Delta t_{\text{total}} = 45.5 - 7.3 = +38.2\mu s$$

- **1 day, the GPS satellite is $38.2\mu s$ ahead of the Earth clock!**

WHAT HAPPENS?

- 1 day, the clock is $38.2 \mu\text{s}$ ahead
- 35 days, the distance is about 400 km (Tokyo-Osaka distance)
- **If you're heading to Tokyo from Tokyo, you'll end up in Osaka!**



HOW ARE WE CORRECTING IT?

- **1 day, we intentionally slow down the clock by $38.2 \mu\text{s}$!**
 - Resulting in a balance
 - Strictly, we use clocks with different clock frequencies
- Theoretical calculation is elegant
- **But the solution is surprisingly simple!**
- GNSS is a combination of elegance and simplicity!

SUMMARY

- GPS measures the time it takes for light to reach from satellites to measuring devices
 - 1 satellite: sphere surface, 2 satellites: circle circumference, 3 satellites: 2 points, 4 satellites: 1 point, and the more satellites, the more we can narrow down the position
- Accurate positioning requires accurate time correction
 - Time delay due to special relativity
 - Time advance due to general relativity
- **Relativity is everywhere in our lives!**

REFERENCES

- International Committee on Global Navigation Satellite Systems, [GNSS, How it Works and Applications](#)
 - Detailed explanation of why positioning becomes more accurate by increasing the number of satellites
 - Many illustrations, easy to understand

REFERENCES

- Neil Ashby, [Relativity in the Global Positioning System](#), DOI: 10.12942/lrr-2003-1
 - We mainly referred to this survey paper for the method of correcting GNSS clocks due to relativistic effects
 - More advanced topics such as geoid correction are explained in detail

REFERENCES

- 内山龍雄, 相対性理論, 岩波書店 (1989)
 - We referred to this textbook for the basic concepts of relativity
 - Compact summary of basic knowledge from special relativity to general relativity
 - Famous opening statement
 - "If you can't understand this, you should probably give up on learning relativity."
 - ...I don't think so

LT SPEAKER RECRUITMENT

- We're recruiting LT speakers at the physics meeting!
 - Any genre is OK!
 - If there's no application, the organizer will hold a Jaiyan Ressairoku (LT event) under the guise of LT...
- If you're interested, join the Discord server of the physics meeting!



ANNOUNCEMENT

- Next meeting is scheduled for May 3
- We're all going to watch the physics video together
- We're also looking for proposals for "I want to watch this video with everyone!"